



**WELCOME
&
THANK YOU**



**<Creative
Software/>**

Docker

For TD Bank



MEET A FEW OF OUR TEAM MEMBERS



TANGY F.
CEO



LINA G
PROJECT MANAGER



CHEDDAE G.
TECH ASSISTANT

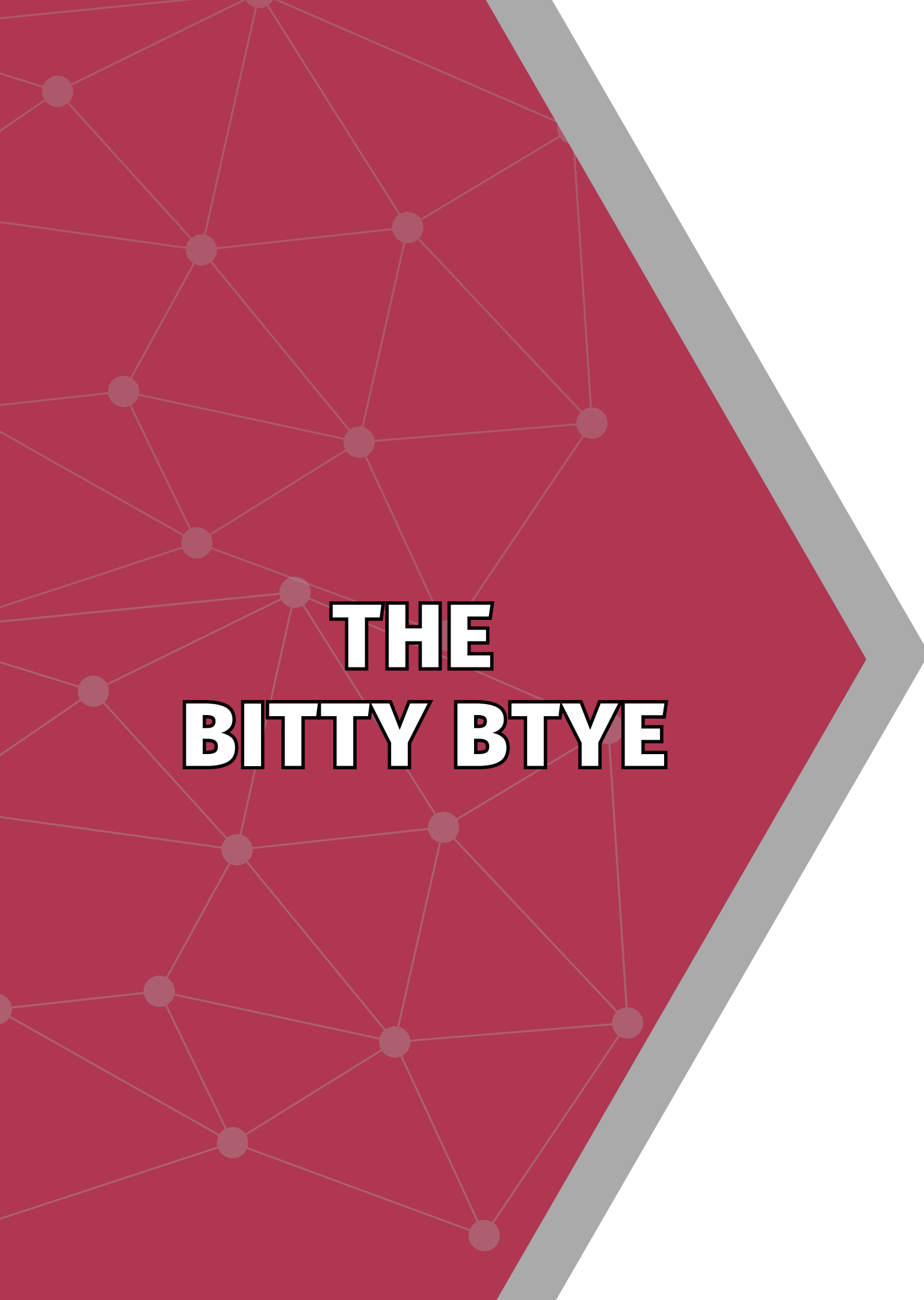


EDDIE K.
DEVELOPER



WILLIAM D.
DEVELOPER

- ***Founded in South Florida in 2018***
- ***Evolved from Software Development Expanded into a Technical Training firm Providing tailored technical training solutions for businesses***



THE BITTY BYTE

**Bitty Byte is our
bite size lighting training**

**The Duration is 1 hour a day
for 5 days**

**One Topic~
In this case it's Docker**

**<Creative
Software/>** Devs.

**WE LISTENED TO
YOU & YOUR
COLLEAGUES**

- ① **We interviewed practitioners and your leads**
- ② **We reviewed anonymous data**
- ③ **Came up with a tailored curriculum based on real-time data**

**<Creative
Software/>** Devs.



THE SPACE

**A safe space where our goal
is to help you**

**We are here to help you
grow. If there is something
you don't understand this is
the place to ask.**

This is a 'No Judgment Zone'

**<Creative
Software/>** 

MEET YOUR BITTY BYTE TEAM



TANGY F.
CEO

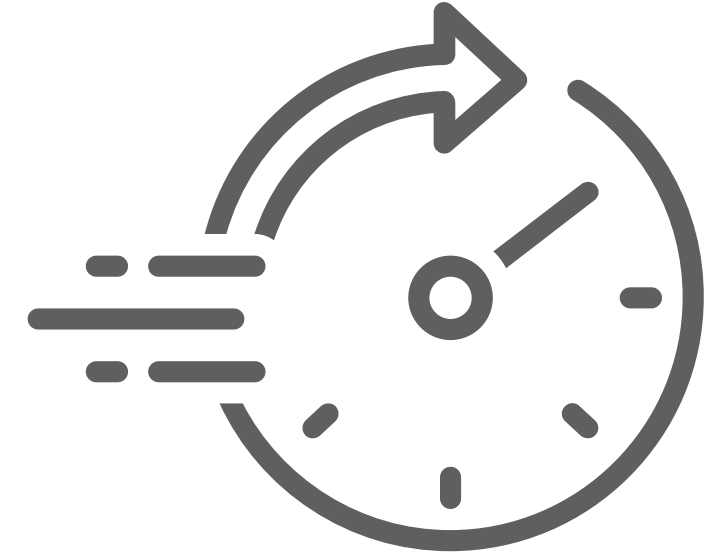


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DEVELOPER

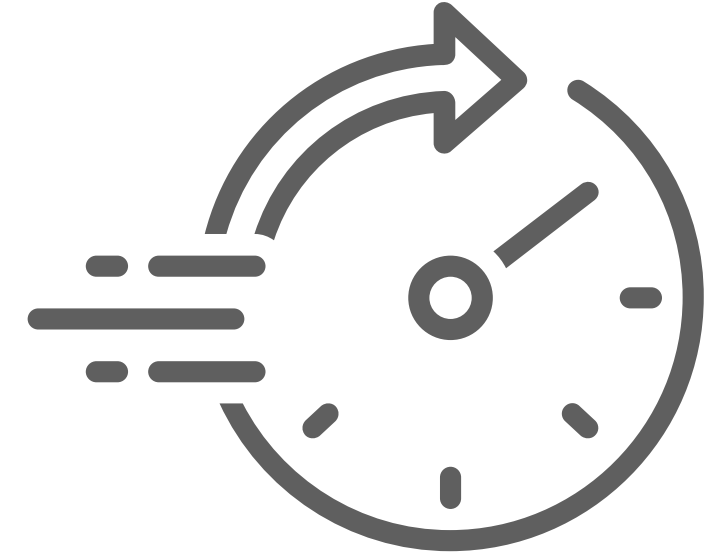
Rapid **TRIVIA**



Rapid Trivia

In what year was Docker officially founded as an open-source project?

Rapid **TRIVIA**



Rapid Trivia

Docker was officially founded as an open-source project in March 2013.

TRAINING DAY 1

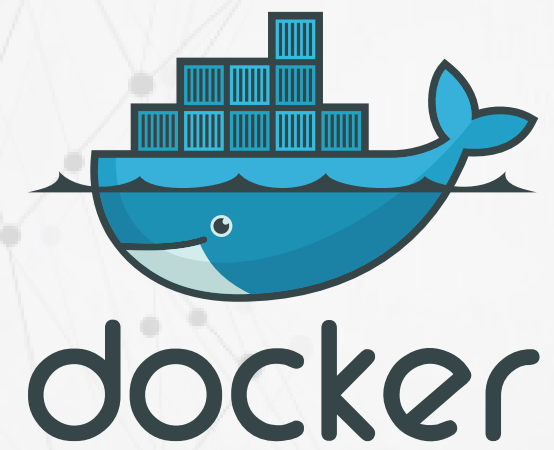
TODAY'S AGENDA

- 1 What is Docker
- 2 Pre Course Required Apps
- 3 Docker Vs. VM's
- 4 Files, Images & Containers
- 5 Starting the Java Project

DAY 1

What is Docker?

WHAT IS DOCKER?



Docker is a software platform that allows you to:

1.Build

2.Test

3.Deploy

Applications quickly.



Docker uses Containers that have everything the software needs to run including libraries, system tools, code & runtime.

DAY 1 GETTING STARTED

Required Apps

Note:
This part is not required on your company laptop.

However,
it is needed to understand the future lesson

REQUIRED APPS

1. Selected language compiler or VM (In this case we are using Java)
2. Having your IDE (VS Code)
3. Adding Extensions for VS Code
 - a. Extension Pack for Java
 - b. Spring boot Extension pack
 - c. Lombok
4. Installing Docker
5. Building your application (Hello World).

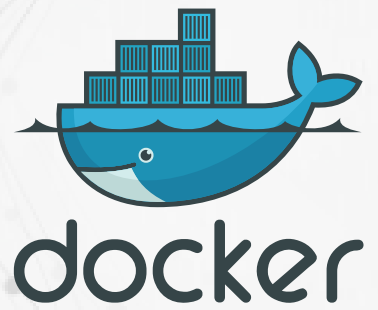


Let's see it running.

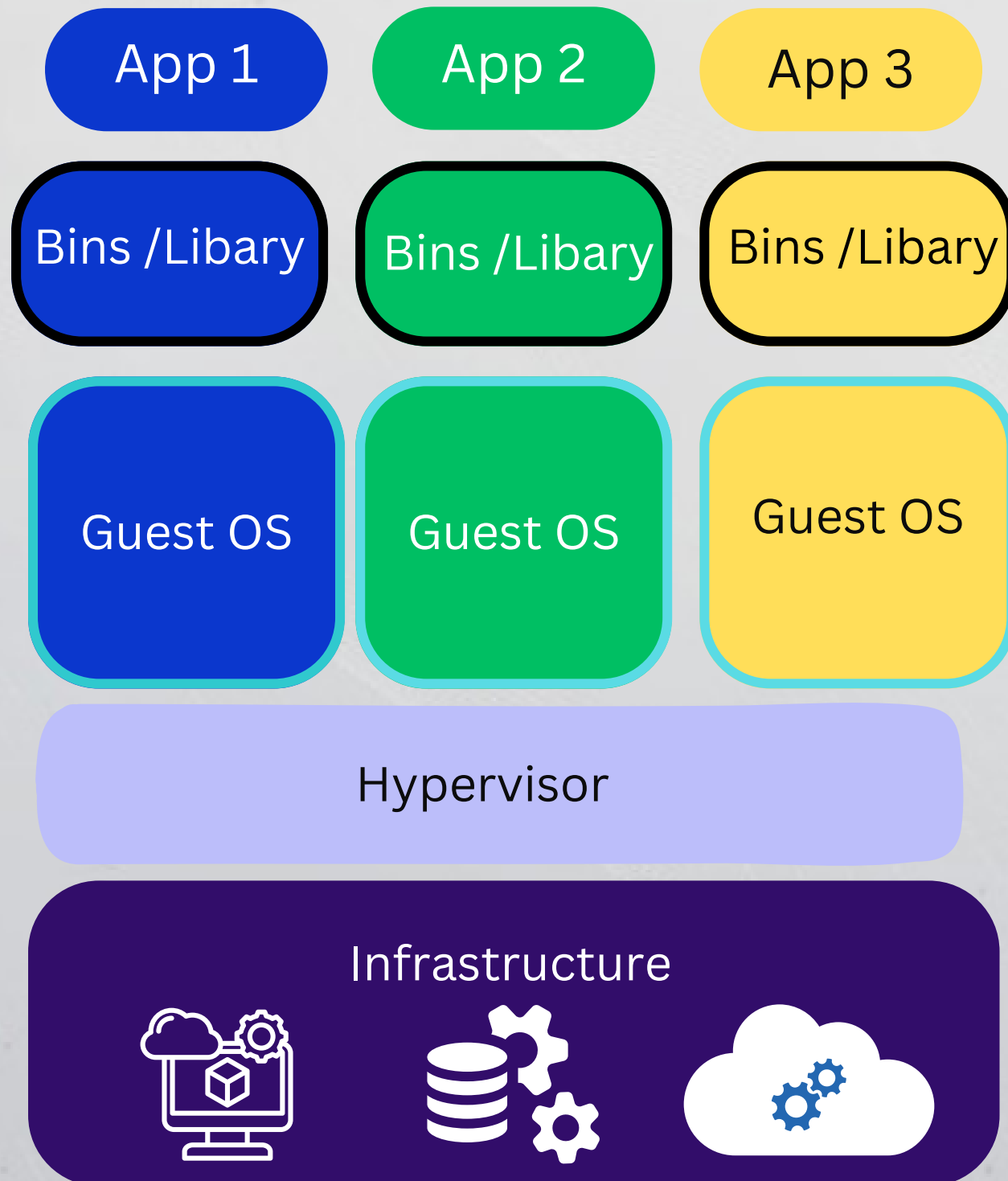
DAY 1

Docker Vs. VM's

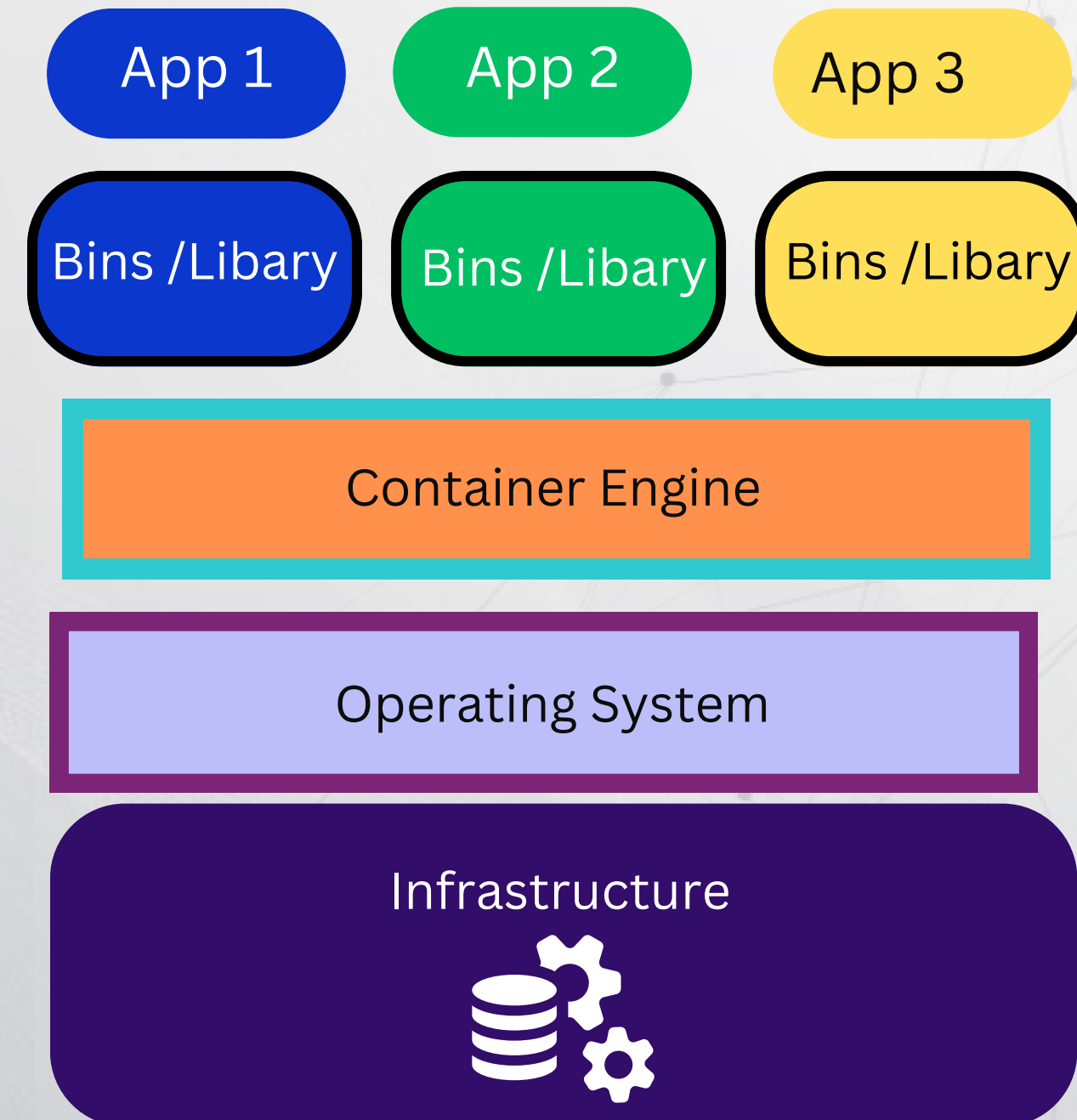
HOW DOCKER DIFFERS FROM VM'S

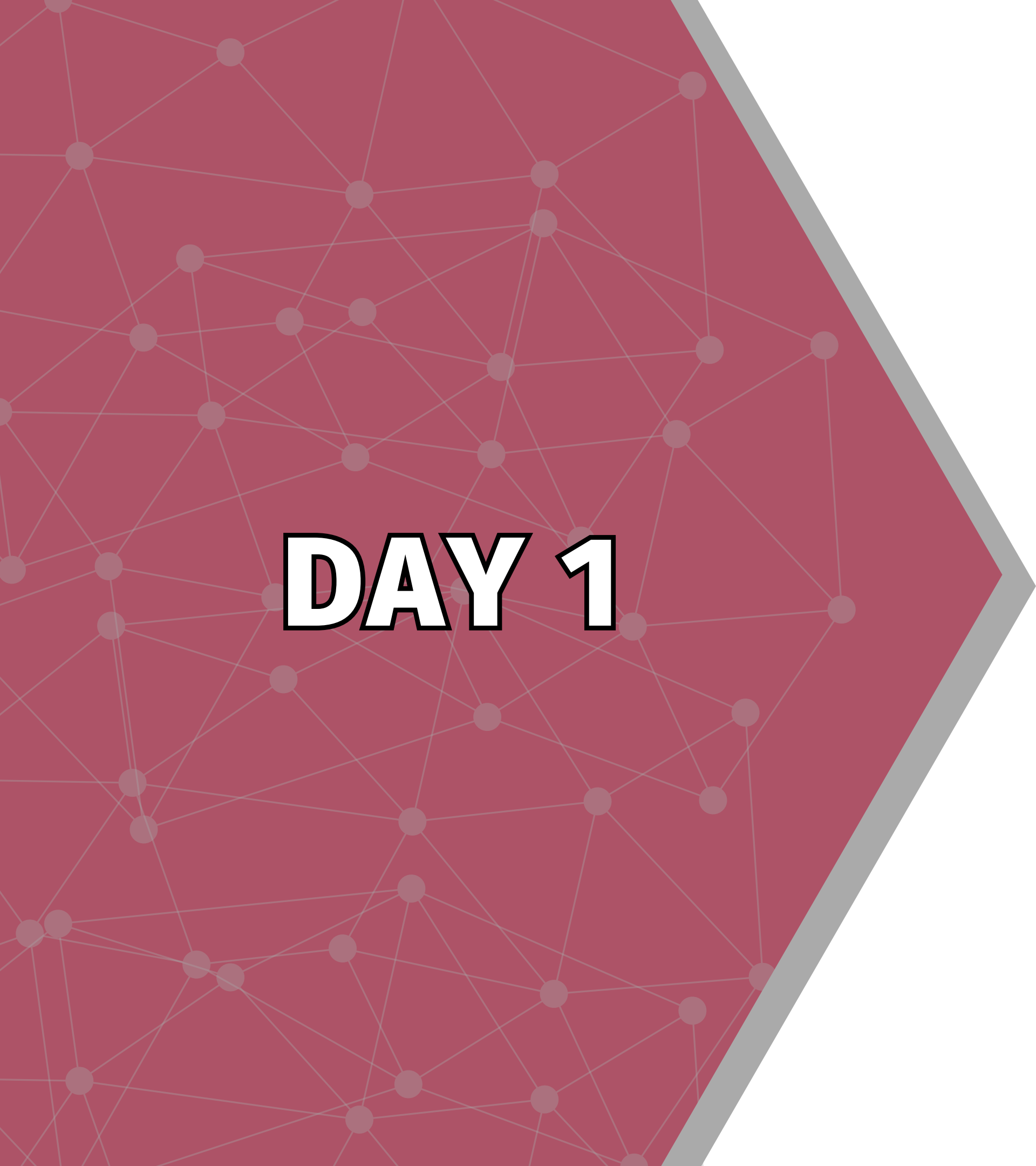


Virtual Machines



Containers



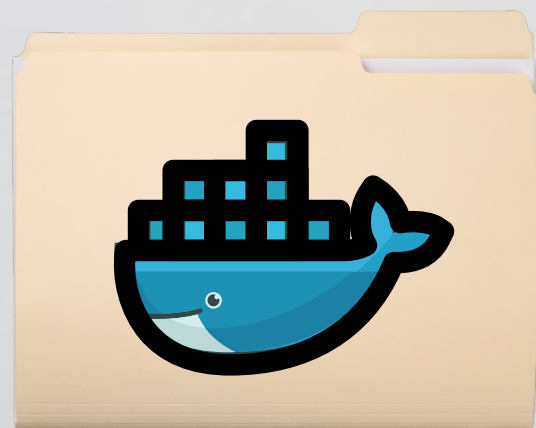


DAY 1

Files, Images & Containers

DOCKER FILE, IMAGE AND CONTAINER

Dockerfile



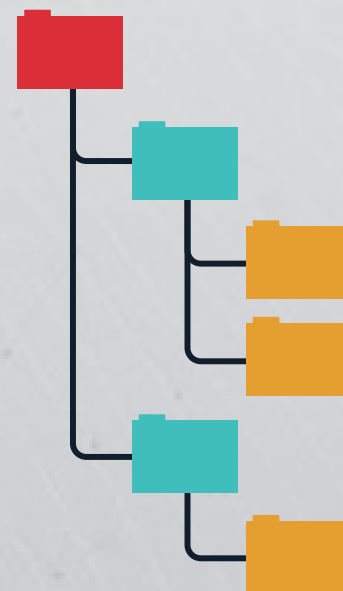
Docker Image



Docker Container



```
FROM openjdk:11-jre-slim
WORKDIR /app
COPY target/your-spring-boot-app.jar app.jar
EXPOSE 8080
CMD ["java", "-jar", "app.jar"]
```



DAY 1

Building your application
(Hello World).

BUILDING YOUR APPLICATION

In VSCode;

Ctrl + Shift + p and type “Spring Initializr” and click:

>Spring Initializr

Spring Initializr: Create a Maven Project...

Then click 3.1.3

Specify Spring Boot version.

3.1.3

Then Java

Specify project language.

Java

Leave default and enter

com.example

Enter spring-docker as Artifact Id

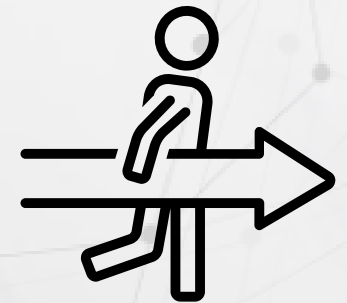
← Spring Initializr: Input Artifact Id

spring-docker

Select jar

Specify packaging type.

Jar



Let's see it running.

BUILDING YOUR APPLICATION

In VSCode;

Then 17:

Specify Java version.

17

Then select lombok, Spring Web and hit enter

Search for dependencies.

☰ Selected 2 dependencies
Press <Enter> to continue.

✓ Lombok	Developer Tools	Selected
Java annotation library which helps to reduce boilerplate code.		
✓ Spring Web	Web	

Then select desired location folder in your local drive and hit enter

Click Open

📘 Successfully generated. Location: c:\Cre8tive\docker-test\my-artifact

Source: Spring Initializr Java Support (Extension)

Open



Let's see it running.

BUILDING YOUR APPLICATION

Create a Dockerfile and add the following content

```
FROM maven:3.8.5-openjdk-17

WORKDIR /app

COPY pom.xml .

RUN mvn dependency:resolve

COPY src ./src

RUN mvn package

CMD ["java", "-jar", "target/spring-app-0.0.1-SNAPSHOT.jar"]
```



Let's see it running.

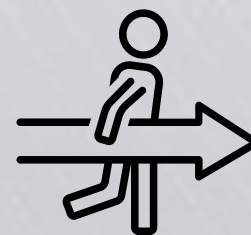
BUILDING YOUR APPLICATION

To create the Image, start Docker Desktop and then run the docker build command in the terminal section of VSCode with the parameters as shown

```
Dockerfile > CMD
1 FROM maven:3.8.5-openjdk-17
2 WORKDIR /app
3 COPY pom.xml .
4 RUN mvn dependency:resolve
5 COPY src ./src
6 RUN mvn package
7 CMD [ "java", "-jar", "target/spring-docker-0.0.1-SNAPSHOT.jar" ]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

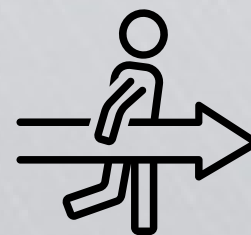
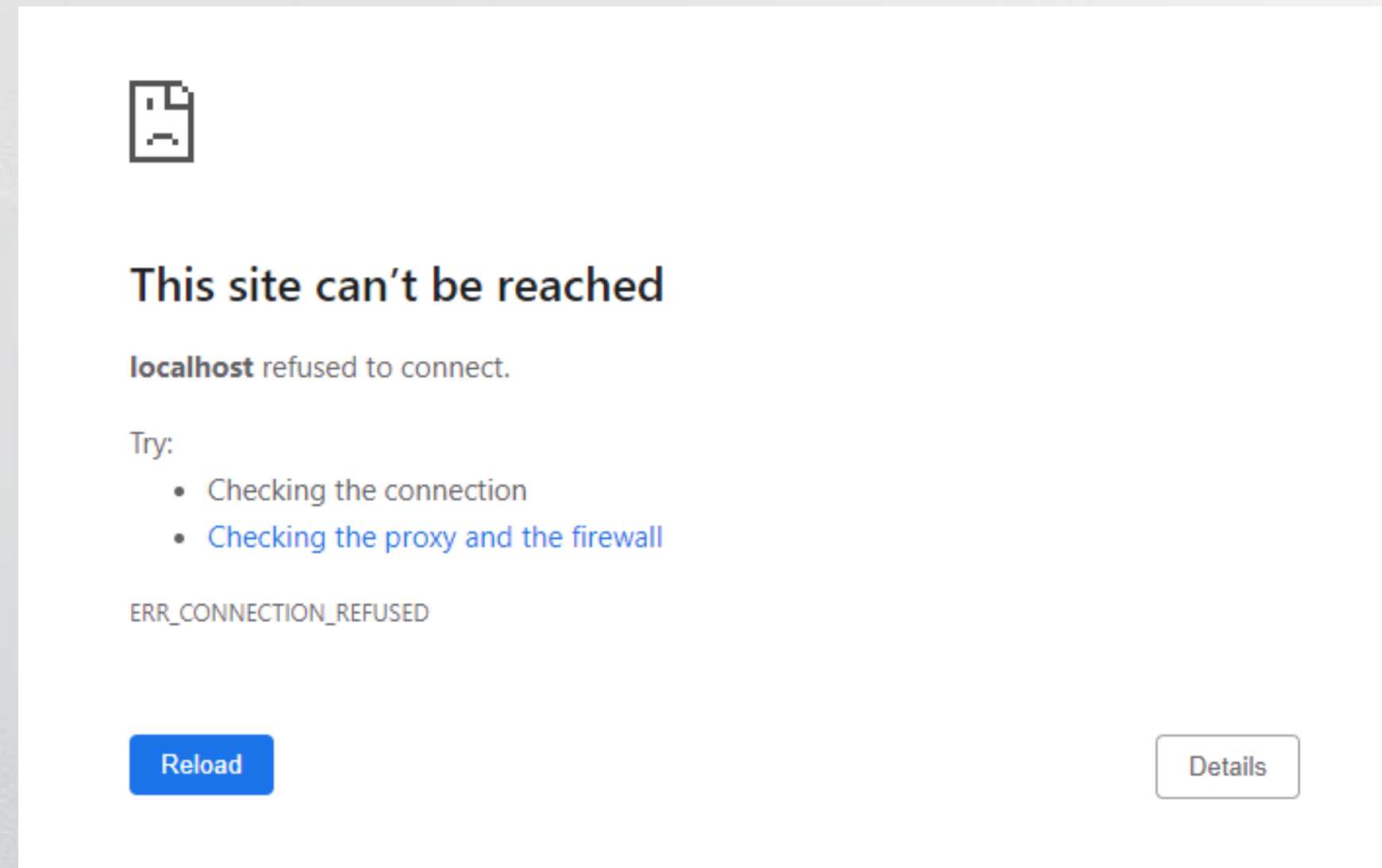
```
PS C:\Cre8tive\docker-test\my-artifact> docker build -t spring-docker-image .
```



Let's see it running.

BUILDING YOUR APPLICATION

After instantiating a Container with the previous image, we are not getting the hello world page in the browser, something is missing.



Let's see it running.

BUILDING YOUR APPLICATION

Add port to expose and re-run docker build command

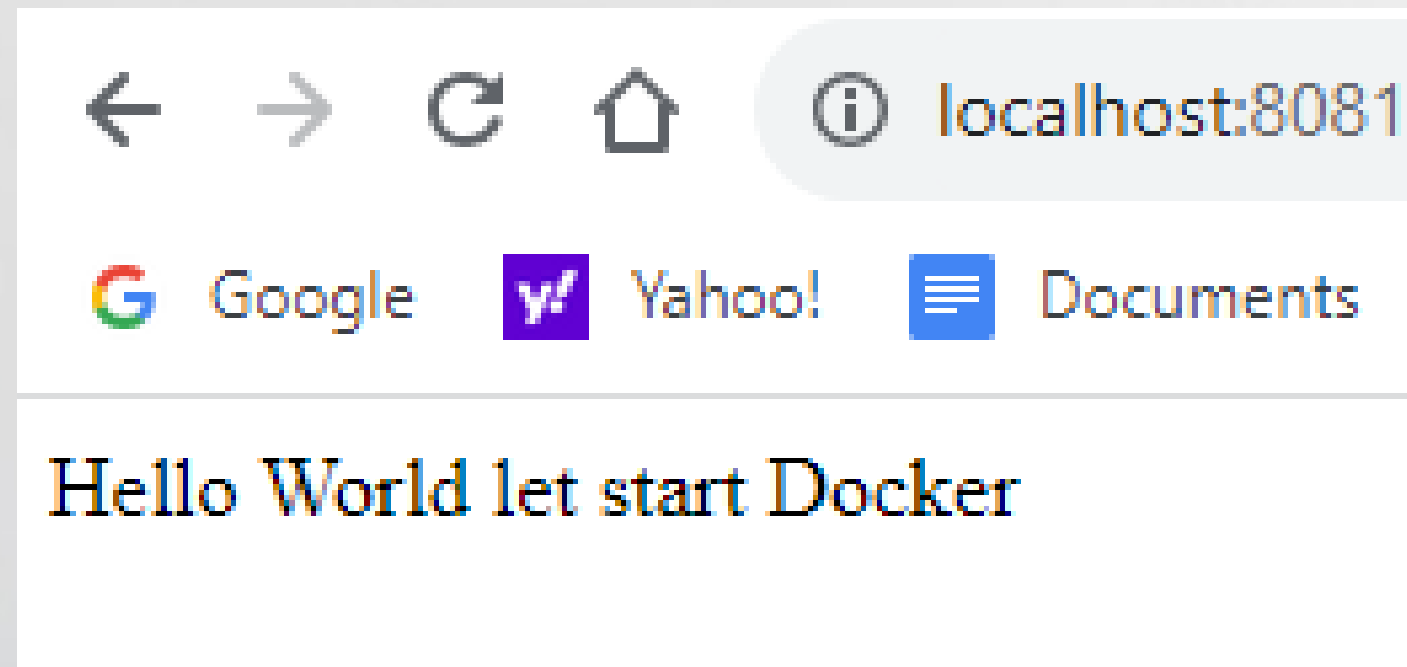
```
Dockerfile X
Dockerfile > ...
1 FROM maven:3.8.5-openjdk-17
2 WORKDIR /app
3 COPY pom.xml .
4 RUN mvn dependency:resolve
5 COPY src ./src
6 RUN mvn package
7 EXPOSE 8080
8 CMD [ "java", "-jar", "target/spring-docker-0.0.1-SNAPSHOT.jar" ]
```



Let's see it running.

BUILDING YOUR APPLICATION

After assigning a port we should get the result



Let's see it running.

TRIVA QUESTION



In Docker:

What is the difference between a Container and an Image?

How does Docker use these concepts to optimize resource usage?



TRIVIA ANSWER



Container vs. Image:

- **An image is a snapshot of an environment for an application, containing all its dependencies.**
- **A container is a running instance of an image.**

Resource Optimization:

- **Docker optimizes resource usage by sharing images and layers.**
- **Containers start quickly and are isolated, ensuring efficient resource utilization.**



CODING EXERCISE



**Create the Hello World in
Spring Boot application and
add it to Docker**





THANK YOU

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Software/>

Bitty Byte

We will see you Thursday

